Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			the *s*-block, *p*-block and *d*-block ✓ the *s*-block and *p*-block only ☐ the *s*-block only ☐ the *d*-block and *p*-block only ☐ the *p*-block only ☐	1			1		
	(b)			titanium / Ti			1	1	1	
	(c)			the Group 1 metals and transition metals all have a +1 oxidation state ☐ the transition metals all have a +3 oxidation state ✓ the Group 1 metals all have a +1 oxidation state ✓ iron and lithium have the same oxidation states ☐ the Group 1 metals and transition metals all have a +4 oxidation state ☐ award (1) for each correct tick if more than two ticks award (1) for two correct and one incorrect award (0) for one correct and two incorrect			2	2		

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<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td></td><td>(d)</td><td></td><td></td><td>agree – award (1) for either of following

• if the compound is coloured it must be a transition
• metal the transition metals have coloured compounds (whereas the Group 1 metals do not)
accept ‘Group 1 metal compounds are white’

assume that coloured means not white

disagree – award (1) for any of following

• titanium has a white compound but is not a Group 1 metal
• titanium <u>also</u> has a white compound
• just because the compound is white does not mean it is in Group 1
neutral answer – some of the transition metal compounds are white</td><td></td><td></td><td>2</td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td>Question 4 total</td><td>1</td><td>0</td><td>5</td><td>6</td><td>1</td><td>0</td></tr>
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